

# G D HRUTHIK

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## PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years of experience in designing and deploying scalable ML pipelines and production-grade models. Proficient in TensorFlow, PyTorch, and CI/CD tools, with proven expertise in containerization (Docker, Kubernetes) and cloud solutions (AWS, Terraform). Demonstrated success in optimizing ML workflows and integrating large language models to drive measurable business impact.

## EDUCATION

**New England College, Henniker, NH**  
*Master of Science, Data Science and Analytics*  
• **GPA:** 3.74

**Jun 2025**

## PROFESSIONAL EXPERIENCE

### Sanofi

**Apr 2024 - Present**

*Machine Learning Engineer*

*Boston, MA*

- Collaborated with clinical teams to establish machine learning pipelines for genomic and clinical trial data, enhancing experiment reproducibility.
- Containerized models using Docker and Kubernetes to enable scalable deployment across oncology and immunology studies.
- Implemented model monitoring, versioning, and compliance checks in line with FDA and GxP standards.
- Contributed to AI-driven internal tools developed with Streamlit and FastAPI, increasing adoption of AI solutions among scientists by 40%.
- Integrated EHR and biomarker data into production pipelines to support translational medicine teams in biomarker discovery.
- Developed a computer vision workflow utilizing CNNs and PyTorch for microscopy image analysis, reducing protein misfolding detection time by 50% compared to manual review.

### Cognizant Technology Services

**Aug 2021 - May 2023**

*Machine Learning Engineer*

*Coimbatore, TN, India*

- Designed real-time anomaly detection models using Autoencoders and Isolation Forests, reducing equipment downtime by 20% for industrial clients.
- Engineered predictive maintenance pipelines integrating Random Forest, Kafka, and Azure Stream Analytics to forecast equipment failures.
- Deployed a fraud detection solution on Azure ML achieving inference speeds under 120ms and enhancing detection accuracy.
- Supported an e-commerce recommendation system leveraging TensorFlow, resulting in a 15% increase in click-through rates during A/B testing.
- Collaborated on end-to-end ML operations pipelines, standardizing containerization and monitoring practices across projects.

## TECHNICAL SKILLS

- **Programming & Data Analysis:** Python, C++, R, SQL, Scala, Pandas, NumPy
- **Machine Learning:** Regression, Classification, Clustering, Dimensionality Reduction, Ensemble Methods (Bagging, Boosting, Stacking), Hyperparameter Tuning (Optuna, Grid Search)
- **Deep Learning & Computer Vision:** CNN, RNN, LSTM, BERT, CodonBert, ProGen2 Transformers, Attention Mechanisms, Reinforcement Learning, CLIP, Image Classification, Object Detection, Fine-tuning Models, Diffusion Models
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, DeepSpeed, Flask, Django
- **Natural Language Processing:** Text Preprocessing, Tokenization, Word Embeddings, Sentiment Analysis, LangChain, RAG, Hugging Face Transformers, OpenAI API
- **Generative AI:** GPT-based LLMs, Diffusion Models, Prompt Engineering
- **Vector Search & Retrieval:** FAISS, Pinecone, Chroma, Weaviate
- **MLOps & Model Deployment:** MLflow, DVC, Docker, Kubernetes, Azure Kubernetes Service (AKS), Streamlit, Ray, TorchServe, FastAPI, Triton Inference Server, TensorFlow Serving
- **CI/CD & Workflow Orchestration:** Jenkins, GitHub Actions, Apache Airflow, Azure DevOps, Luigi
- **Cloud Platforms:** AWS SageMaker, GCP Vertex AI, Azure ML Studio
- **Visualization & Reporting:** Tableau, Matplotlib, Seaborn
- **Data Engineering:** Hadoop, PySpark, Azure Databricks, Data Factory, Synapse Analytics, Snowflake
- **Project & Collaboration Tools:** Jupyter, Google Colab, VS Code, Agile, A/B Testing, Requirement Gathering